Algorithm Design Techniques

Introduction: The Brute Force

Algorithm Design

- When searching for a solution to an algorithmic problem, we may be interested in two types:
  - We are looking for the optimal (Exact) solution
  - We are interested in a solution which is *good enough*, where good enough is defined by a set of parameters (Approximately optimal solution).
- In general, it is hard to design optimal algorithms that are:
  - correct
  - efficient
  - implementable

Algorithm Design

- From now on, we are going to study some basic and general strategies in designing algorithms to solve some typical computing problems.
- We will analyze the efficiency of these algorithms using the tools we have seen in the past few lectures.
- We will learn how to design feasible or partial solutions with better efficiency:
  - A *feasible solution* is a solution which satisfies any given requirements
  - A *partial solution* is a solution which is not complete but could possibly be extended/improved.

Algorithm Design Techniques

- Brute Force
- Divide and Conquer
- Greedy Algorithms
- Dynamic Programming
- Transform and Conquer
- Backtracking
- Genetic Algorithms
- ……..

Brute Force

“Et tu, Brute?”

“You too, Brutus?“

-Julius Caesar's last words

Brute Force Strategy for Algorithm Design

- Brute Force is a straightforward approach to solving a problem
- The strategy is directly based on the problem's statement and definitions.
- But ,all possible solutions have to be considered , this often takes too much time to run.
- In many cases, Brute Force does not provide us very efficient solutions.

Brute Force : Some Examples

Solutions are based on the problem's statement and definition :

- Exponentiation (standard algorithm)
- Calculating a polynomial (standard algorithm)
- Matrix multiplication (standard algorithm)
- Sequential search
- Sum of an array (standard algorithm)
- Max /min of an array (standard algorithm)
- Simple sorting methods
- Exhaustive search
  - TSP problem
  - Knapsack problem
  - Assignment problem
- Exhaustive search in graphs
  - BFS, DFS, …
- …

Brute Force Search and Sort

- Remember that , in simple search and sort algorithms the entire list have to be scanned.
- These algorithms are effectively brute force algorithms. Because,in the worst case we have to check every possibility.
- Worst case complexities of some algorithms in this category are :
  - Sequential Search $O(n)$
  - Selection Sort $O(n^2)$
  - Insertion Sort $O(n^2)$
  - Bubble Sort $O(n^2)$
  - ……

  First,we consider two simple problems:1.Exponentiation,2.Evaluation of polynomials.

Brute Force Exponentiation : Iterative solution

Compute nth power of positive a : $a^n = a \times a \times \ldots\ldots \times a$

// The computation requires one loop for multiplications

```
res = 1
for i=1 to n
        res=res*a
return res
```

Or

```
res = 1
for i=n  to 1
        res=res*a
return res
```

The number of multiplications for both solutions:

$$T(n) = \sum_{i=1}^{n} 1 = n$$

$$= \Theta(n)$$

Recursive Exponentiation

```
Exp(a, n)
 //Computes a^n , n is a non-negative integer
  if n == 0
     return 1
  return (a * exp(a, n- 1) )
```

Complexity : $\Theta(n)$, we have to perform n multiplications

Brute force polynomial evaluation

Problem:

Find the value of a polynomial at a point $x = x_0$

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x^1 + a_0$$

Example :

$$p(x) = 4x^4 + 7x^3 - 2x^2 + 3x^1 + 6$$

Brute force solution: Perform all multiplications in every term:

Computation from LEFT to RIGHT

$$p(x) = 4*x*x*x*x + 7*x*x*x - 2*x*x + 3*x + 6$$

Brute force polynomial evaluation

//The a terms in p(x) are passed in parameter P. (evaluate from LEFT to RIGHT)

$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x^1 + a_0$

Poly1(P, x_0)

```
  x=x0
  p= 0.0
  for i = n down to 0 do
    power = 1
    for  j = 1 to i do
        power =  power * x        //compute x^i
    p = p + a[i] * power
 return p
```

Efficiency : Two nested for loops :→ $T(n)= \Theta(n^2)$

Can we do better? Yes.

Polynomial evaluation: Improvement

We can do better by evaluating from **right to left:**

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x^1 + a_0$$

```
Poly2(P , x0)
x = x0
p = a[0]
power = 1
for i =  1 to n do
   power =  power * x
   p = p + a[i] * power
return p
```

Efficiency : One for loop→ T(n)= Θ(n)

→One order of magnitude improvement!

Brute Force Examples #1 : Selection Sort

- We have discussed Insertion Sort, now we consider Selection Sort:

  - **Scan the entire given list to find its smallest element and swap it with the first unsorted element.**
  - **Repeat for every next element.**

  - This is a straightforward solution : Brute Force strategy

  Time efficiency : $\Theta(n^2)$

  Why ?

  Each time we have to find the minimum by brute force.

  Bubble sort is also a Brute Force algorithm.

Brute Force Examples #1 : Selection Sort

```
SelectionSort(A,n)
    for i = 0 to n-2 do
        min ← i
        for j = i+1 to n-1 do
            if A[j] < A[min]
                min← j
        swap (A[i] , A[min])
```

| 89 45 68 90 29 34 **17**

17 | 45 68 90 **29** 34 89

17 29 | 68 90 45 **34** 89

17 29 34 45 | 90 **68** 89

17 29 34 45 68 | 90 **89**

17 29 34 45 68 89 | 90

Brute Force Examples #1 : Selection Sort, An Analysis

How many times the second loop is executed?

comparisons :

$$T1(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1$$

$$= \sum_{i=0}^{n-2} [(n-1) - (i+1) + 1]$$

$$= \frac{(n-1)n}{2}$$

$\rightarrow T1(n) = \Theta(n^2)$

Brute Force Examples #2: Closest-Pair of Points

- Find the two closest points in a set of n points on a plane
- Points can be airplanes (most probable collision candidates), database records, DNA sequences, etc.
- Cluster analysis: pick two points, if they are close enough they are in the same cluster, pick another point, and so on…

Brute Force Examples #2: Closest-Pair of Points

- For simplicity we consider 2-D case
- Euclidean distance :

$$d(p_i, p_j) = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}$$

- Brute-force: compute distance between each pair of disjoint points and find a pair with the smallest distance.Since

  $d(p_i, p_j) = d(p_j, p_i)$,

  we consider only $d(p_i, p_j)$ for $i < j$

Brute Force Examples #2: Closest-Pair of Points

```
BruteForceClosestPair( P )
//Input: A list P of n (n≥2) points p1(x1,y1), …, pn(xn,yn)
//Output: distance between closest pair
  d ← ∞     //Initially, later it is minimized
  for i ← 1 to n-1 do
        for j ← i+1 to n do
           d ← min( d, sqrt((x_i − x_j)^2 + (y_i − y_j)^2 ) )
  return d
```

Brute Force Examples #2: Closest-Pair of Points, An Analysis

Input size: n

- Basic operation: Computing square root → Costly
- Actually , computing the square root is not needed . The result will be the same if we consider only the squares:
- We will find the same pair of points in both cases. So,the basic operation can be taken as squaring two numbers:

  How many times?

  $T(n) = \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} 2 = 2 \sum_{i=1}^{n-1} (n - i)$

  $= 2[(n-1)+(n-2)+\ldots+1]$

  $= 2\sum_{i=1}^{n-1} i = 2\,[n(n+1)/2 - n]$

  $= (n-1)n$

  $= \Theta(n^2)$

Brute Force Examples #3: Exhaustive Search

State-space search:

- Given an initial state, a goal state and a set of operations,
- Find a sequence of operations that transforms the initial state to the goal state.

The solution process can be represented as a tree

Brute Force Examples #3: Exhaustive Search

- Examples: Brute force solution of
  - Traveling salesman problem
  - Knapsack problem
  - Assignment problem
  - Cryptography

We are going to consider some introductory examples.

(Better solutions will be considered later)

Exhaustive Search: Traveling Salesman Problem (TSP)

- Find the shortest tour through a given set of n cities that visits each city exactly once before returning to the city where it started
- Can be conveniently modeled by a weighted graph; vertices are cities and edge weights are distances
- Same as finding a "Hamiltonian Circuit" in a graph:

→A circuit is a path with no repeating edges that begins and ends at the same vertex.

A circuit that doesn't repeat vertices is called a cycle.

Hamiltonian circuits and TSP

- Hamiltonian path: A path that uses each vertex of a graph exactly once.
- Hamiltonian Circuit: If the path ends at the starting vertex, it is called a **Hamiltonian circuit.**

→A sequence of n+1 adjacent vertices :

$$v_{i_0}, v_{i_1}, \ldots, v_{i_{n-1}}, v_{i_0}.$$

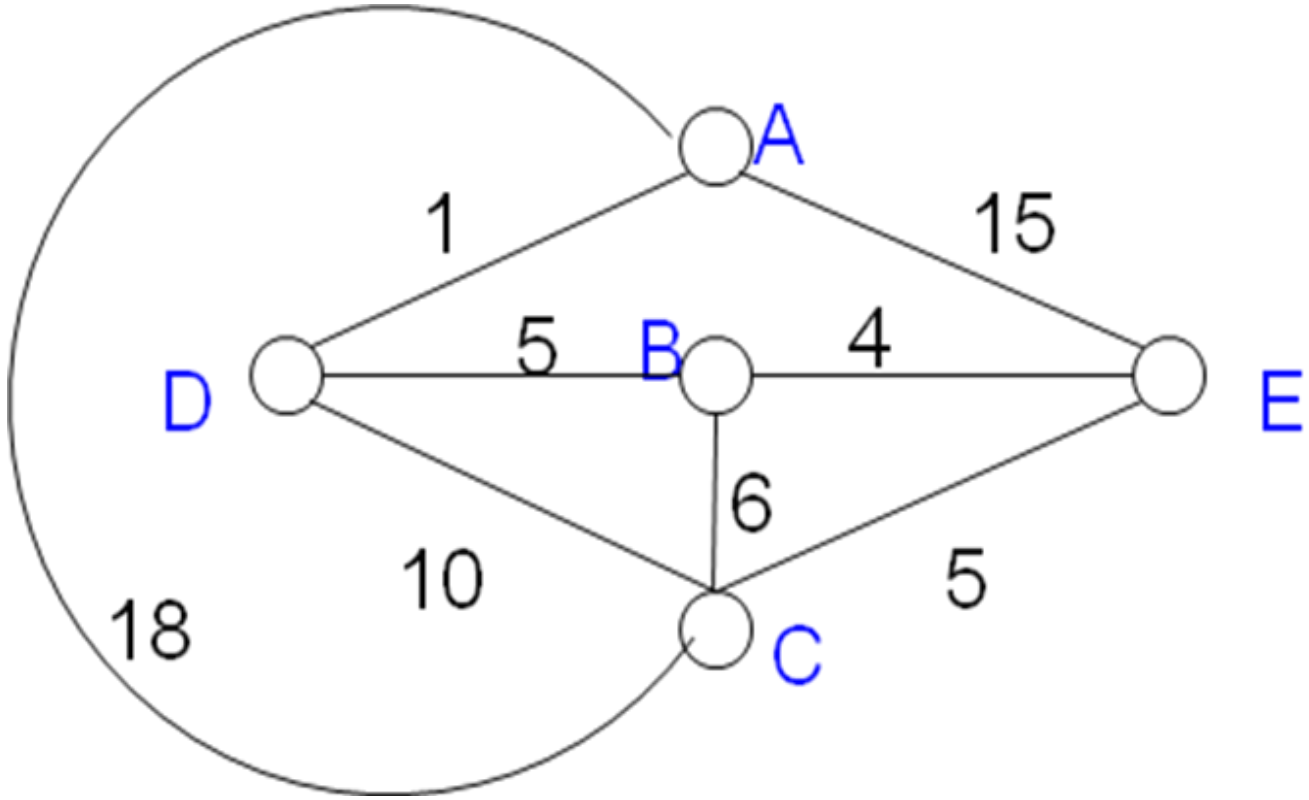
How can we solve TSP by Brute Force?

Traveling Salesperson Problem

- A salesperson has a list of cities, each of which (s)he must visit *exactly once* before returning to the initial city.
- There are direct roads between some pairs of cities as shown on a map.
- Find the route the salesperson should follow for the shortest possible round trip that both starts and finishes at the same given home city of the salesperson.

TSP : Illustration

Find the shortest TSP tour starting at A :



Hamiltonian Circuits and TSP

Given a directed graph G = (V , E):

city → vertex, road → edge, length of the road → edge weight.

TSP → Find a shortest Hamiltonian Circuit : A cycle that passes through all the vertices of the graph exactly once.

- Exhaustive search by Brute Force:
  - List all the possible Hamiltonian circuits (starting from any vertex)
  - Ignore the direction
  - How many candidate circuits do we have?

TSP :Complexity

paths =(n-1)! possible paths for a directed graph.

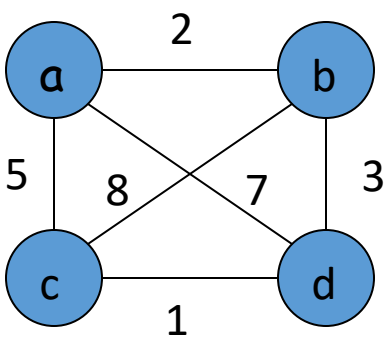
= (n-1)!/2 “ “ undirected

= n! possible paths if home city is not fixed

In all cases we have a $\Theta(n!)$ problem !

intractable! Why ?

TSP Solution by Exhaustive Search : Example



All permutations of cities starting at a

$\frac{1}{2}$(n-1)! permutations

a → b → c → d → a	2+8+1+7 = 18
a → b → d → c → a	2+3+1+5 = 11 ← **optimal**
a → c → b → d → a	5+8+3+7 = 23
a → c → d → b → a	5+1+3+2 = 11 ← **optimal**
a → d → b → c → a	7+3+8+5 = 23
a → d → c → b → a	7+1+8+2 = 18

Brute Force : Cryptography

- In cryptography, a brute-force attack consists of an attacker trying many passwords with the hope of eventually guessing the password correctly.
- The attacker systematically checks all possible passwords and passphrases until the correct one is found.
- As the password's length increases, the amount of time to find the correct password increases exponentially.
- The resources required for a brute-force attack grow exponentially with increasing key size.
- Modern symmetric algorithms typically use computationally stronger 128- to 256-bit keys which are (almost)impossible to crack.

Brute Force Cryptograpy :Example

- Lets say if we have an alphanumeric 8-character password.

  →We can have 52 possible letters (English Alphabet)
- If we add the Numeric digits,we have 62 characters in total.
- Brute force will check all possible passwords:
  - For 8-character-password, the number of possible passwords is : 62^8= 218.340.105.584.896
- Assume our system can produce 1 result per second.

  → Checking all possibilities would take 218 trillion seconds

  → ~7 million years would be needed to crack the password.

Warning: Current day hardware and software for fast computing can reduce this time to just a few seconds!

Brute-Force Strengths

- Simplicity and wide applicability
- Yields reasonable algorithms for some important problems such as : matrix multiplication, sorting, searching, string matching…
- Also yields standard algorithms for simple computational tasks and graph traversal problems
- Brute-force techniques are inefficient, but we may use them to evaluate solutions found through other algorithms.
- On the other hand, Brute Force may be feasible for moderate size problems with the increased powers of current computers.

Brute-Force Weaknesses

- Rarely yields efficient algorithms
- Some brute-force algorithms are unacceptably slow
- Not as constructive and elegant as some other design techniques.
- In general ,Brute Force is the naivest way to search for solution. [to any real-life problem ?].